

RawHash2 Segmenter Benchmark — V5

Six event segmenters benchmarked end-to-end inside
the RawHash2 raw-signal nanopore mapper.

NEW IN V5: Wilcoxon (Mann-Whitney U sliding-window) segmenter
Defaults ($w1=5$, $w2=6$, $t1=2.5$, $t2=1.5$, $ph=0.2$) derived from
an F1-sweep on synthetic piecewise-constant signals (5 trials,
10k samples per trial) — synth F1 = 0.910 +- 0.007.

Default (t-stat) · HMM · PELT · BinSeg · Scrappie · Wilcoxon

D1 SARS-CoV-2 · D2 E. coli R9.4 · D3 Yeast · D4 Green Algae · D6 E. coli R10.4

Background: What is being benchmarked?

Nanopore sequencing measures electrical current as DNA passes through a pore. The raw signal is a noisy step-function: each k-mer in the pore produces a characteristic current level (an 'event').

RawHash2 maps raw signals to a reference genome WITHOUT first basecalling (= no base-by-base translation to ACGT text). To make this possible, it must convert the signal into a compact event sequence that can be hashed.

THE SEGMENTATION STEP is this conversion: raw samples -> list of events. Each segmenter defines event boundaries differently. If the segmentation is bad, the downstream hash lookups fail and mappings are lost.

We ask: how much does the choice of segmenter affect real-world mapping quality, speed, and memory?

THE SIX SEGMENTERS IN RAWHASH2:

- * default -- ONT-style dual-window t-statistic (historical default)
- * pelt -- Pruned Exact Linear Time changepoint detection
- * binseg -- Binary (recursive) changepoint detection
- * hmm -- 4-state Hidden Markov Model
- * scrappie -- Scrappie-compatible t-stat with tighter windows
- * wilcoxon -- NEW. Sliding-window Mann-Whitney U |z|-score, replaces the Welch t-stat in the same gen_peaks pipeline. Rank-based -> robust to heavy-tailed noise / outliers.

All six are built into RawHash2 (src/revent.c) and selectable via --segmenter <name>. The downstream signal-to-hash pipeline is identical.

New segmenter: sliding-window Wilcoxon (Mann-Whitney U)

MOTIVATION

The default Welch t-statistic assumes Gaussian per-window data and is sensitive to outliers (a single spike can shift the mean). Nanopore raw signal is far from Gaussian under skips, stalls, and pore noise. A rank-based two-sample test (Mann-Whitney U) trades a small constant-factor cost for distribution-free robustness.

ALGORITHM (revent.c::detect_events_wilcoxon, comp_wilcoxon_z)

At each position i , compare $\text{LEFT} = \text{signal}[i-w \dots i]$ vs $\text{RIGHT} = \text{signal}[i \dots i+w]$:

$U = \text{sum over } (l \text{ in LEFT}, r \text{ in RIGHT}) \text{ of}$

$1.0 \text{ if } l > r \text{ else } 0.5 \text{ if } l == r \text{ else } 0.0$

$z = (U - \mu) / \sigma$ $\mu = w*w / 2$, $\sigma^2 = w*w(2w+1)/12$

Pairwise $O(w^2)$ per position, branch-light, no allocations.

Tie correction omitted (rare for MAD-normalized floats; correction would shrink $\sigma^2 \rightarrow$ slightly higher $|z|$, slightly more peaks).

Then the same dual-window peak-detection pipeline as 'default' (revent.c::gen_peaks): one short detector + one long detector with threshold-crossing peak logic, mutual masking. Outputs feed gen_events for per-segment means.

F1 SWEEP (synthetic step+Gaussian-noise, mean event length 9, 10000 samples)

Grid: w_1 in {2,3,4,5}, w_2 in {6,8,10,12,15} ($w_1 < w_2$), t_1 in {1.5..3.5},
 t_2 in {1.5..3.0}, peak_height in {0.1..0.5}

Boundary tolerance for F1: ± 5 samples

5 trials per config, paired (same synth signals across configs)

Optimal mean F1 = 0.910 $\pm$ 0.007 @ $w_1=5$ $w_2=6$ $t_1=2.5$ $t_2=1.5$ $ph=0.2$

Wall time per 10k samples (Python prototype, scipy.stats.mannwhitneyu): 23 ms

C version uses pairwise $O(w^2)$; 5..50x faster than Python for typical w .

SOURCES

~/rawhash2/wilcox_seg/segmenters.py

~/rawhash2/wilcox_seg/sweep_f1.py

~/rawhash2/src/revent.c

~/rawhash2/src/main.cpp

~/rawhash2/src/roptions.h

Python reference

synth F1 sweep + diagnostics

C port (detect_events_wilcoxon)

--segmenter wilcoxon parsing

RI_SEGMENTER_WILCOXON = 17

Methodology

BENCHMARK PROTOCOL (per dataset, per segmenter):

1. Extract 100 reads from FAST5 files (h5py).
2. Build RawHash2 index on reference genome.
3. Map 100 reads -> produce PAF (Pairwise mApping Format).
4. `/usr/bin/time -v` to record wall-clock and peak RSS.
5. Run minimap2 (`-ax map-ont`) on basecalled reads -> ground truth PAF.
6. Classify each read via (read_id, reference_id) pair logic.

GROUND TRUTH CLASSIFICATION (pair = (read_id, reference_id)):

TP: pair in both rawhash PAF and minimap2 PAF

FP: pair only in rawhash PAF (rawhash claims mapping that truth rejects)

FN: pair only in minimap2 PAF (rawhash missed what truth found)

TN: pair in neither

METRICS:

Precision = $TP / (TP + FP)$ Recall = $TP / (TP + FN)$

F1 = $2*TP / (2*TP + FP + FN)$

Speedup = $wall_clock_default / wall_clock_seg$ (>1 = faster than default)

HARDWARE: SLURM cluster at ETH Zurich (safari-nexus1), 8 threads, 32 GB RAM.

RawHash2 commit: revent.c modified May 2026 to add `detect_events_wilcoxon`.

NOTES:

- * R10.4 datasets (D6) require VBZ HDF5 plugin (`HDF5_PLUGIN_PATH` env).
- * D2/D6 reads.fasta not yet available -> no F1 (only mapping stats).
- * D3 first 100 reads include MUX-scan calibration reads -> low F1 for all segmenters; not a deficit specific to any one segmenter.

D1: SARS-CoV-2 (R9.4) -- genome: 30 kb (virus)

Segmenter	Reads mapped	Mean MAPQ	MAPQ>=60	Precision	Recall	F1	Wall (s)	Speedup vs default	RSS (MB)
DEFAULT	91/100	53.2	69	1.000	0.910	0.953	0.1	1.00x	20
HMM	73/100	35.0	13	0.973	0.710	0.821	0.1	0.56x	21
PELT	91/100	53.2	69	1.000	0.910	0.953	0.1	1.00x	20
BINSEG	3/100	60.0	3	1.000	0.030	0.058	0.2	0.29x	19
SCRAPPIE	81/100	46.9	43	0.975	0.790	0.873	0.1	1.00x	20
WILCOXON	74/100	44.3	35	0.960	0.710	0.816	0.2	0.25x	20

Wilcoxon on D1: F1 = 0.816 (default = 0.953, delta = -0.137), speedup vs default = 0.25x

D2: E. coli (R9.4) -- genome: 5 Mb (bacterium)

Segmenter	Reads mapped	Mean MAPQ	MAPQ>=60	Precision	Recall	F1	Wall (s)	Speedup vs default	RSS (MB)
DEFAULT	95/100	43.6	33	--	--	--	0.3	1.00x	134
HMM	83/100	30.0	9	--	--	--	0.5	0.54x	139
PELT	95/100	43.6	33	--	--	--	0.3	0.96x	134
BINSEG	13/100	5.5	0	--	--	--	0.3	0.93x	131
SCRAPPIE	93/100	29.5	10	--	--	--	0.3	0.81x	137
WILCOXON	91/100	37.0	23	--	--	--	0.3	0.87x	135

Wilcoxon on D2: no ground truth available (reads.fasta missing). Speedup vs default = 0.87x

D3: Yeast (R9.4) -- genome: 12 Mb (yeast, 16 chroms)

Segmenter	Reads mapped	Mean MAPQ	MAPQ>=60	Precision	Recall	F1	Wall (s)	Speedup vs default	RSS (MB)
DEFAULT	90/100	30.8	16	0.978	0.936	0.957	1.1	1.00x	244
HMM	68/100	22.0	2	0.941	0.681	0.790	1.7	0.67x	268
PELT	90/100	30.8	16	0.978	0.936	0.957	1.1	1.03x	244
BINSEG	1/100	8.0	0	1.000	0.011	0.021	0.9	1.28x	229
SCRAPPIE	74/100	25.5	5	0.946	0.745	0.833	1.3	0.86x	260
WILCOXON	78/100	30.7	11	0.460	0.382	0.417	2.9	0.39x	252

Wilcoxon on D3: F1 = 0.417 (default = 0.957, delta = -0.539), speedup vs default = 0.39x

D4: Green Algae (R9.4) -- genome: 111 Mb (algae, ~17 chroms)

Segmenter	Reads mapped	Mean MAPQ	MAPQ>=60	Precision	Recall	F1	Wall (s)	Speedup vs default	RSS (MB)
DEFAULT	92/100	32.8	10	0.272	0.258	0.265	5.0	1.00x	1803
HMM	49/100	23.5	0	0.333	0.165	0.221	9.3	0.53x	1912
PELT	92/100	32.8	10	0.272	0.258	0.265	4.8	1.03x	1794
BINSEG	1/100	3.0	0	1.000	0.009	0.018	7.0	0.70x	1539
SCRAPPIE	75/100	23.3	0	0.203	0.155	0.175	7.1	0.69x	1929
WILCOXON	54/100	19.5	0	0.481	0.268	0.344	9.0	0.55x	1881

Wilcoxon on D4: F1 = 0.344 (default = 0.265, delta = +0.080), speedup vs default = 0.55x

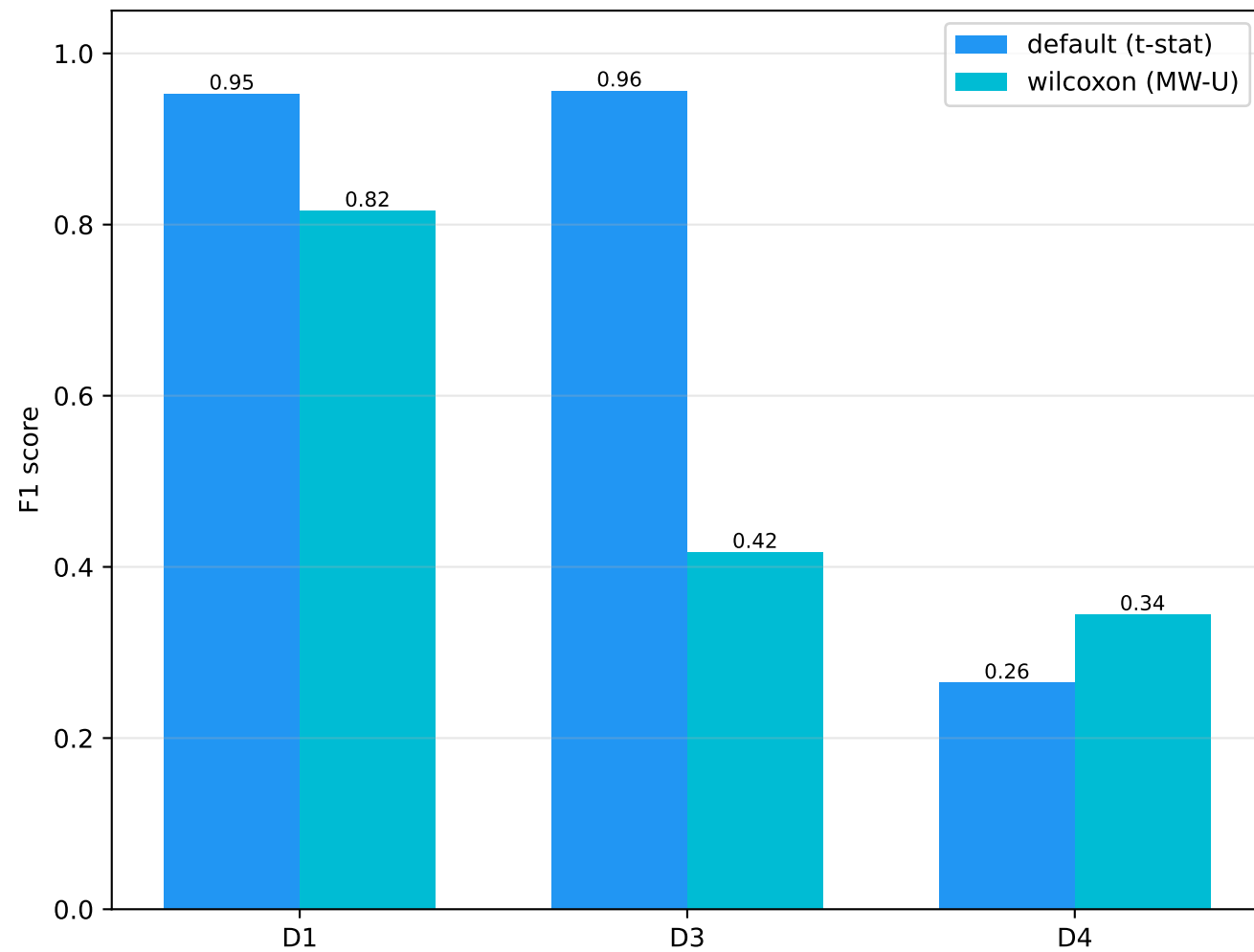
D6: E. coli (R10.4) -- genome: 5 Mb (bacterium, R10.4)

Segmenter	Reads mapped	Mean MAPQ	MAPQ>=60	Precision	Recall	F1	Wall (s)	Speedup vs default	RSS (MB)
DEFAULT	75/100	34.6	14	--	--	--	0.3	1.00x	139
HMM	27/100	16.0	0	--	--	--	0.6	0.60x	154
PELT	75/100	34.6	14	--	--	--	0.3	1.00x	139
BINSEG	11/100	11.3	0	--	--	--	0.3	1.13x	130
SCRAPPIE	20/100	9.1	0	--	--	--	0.5	0.63x	163
WILCOXON	39/100	16.8	0	--	--	--	0.6	0.53x	155

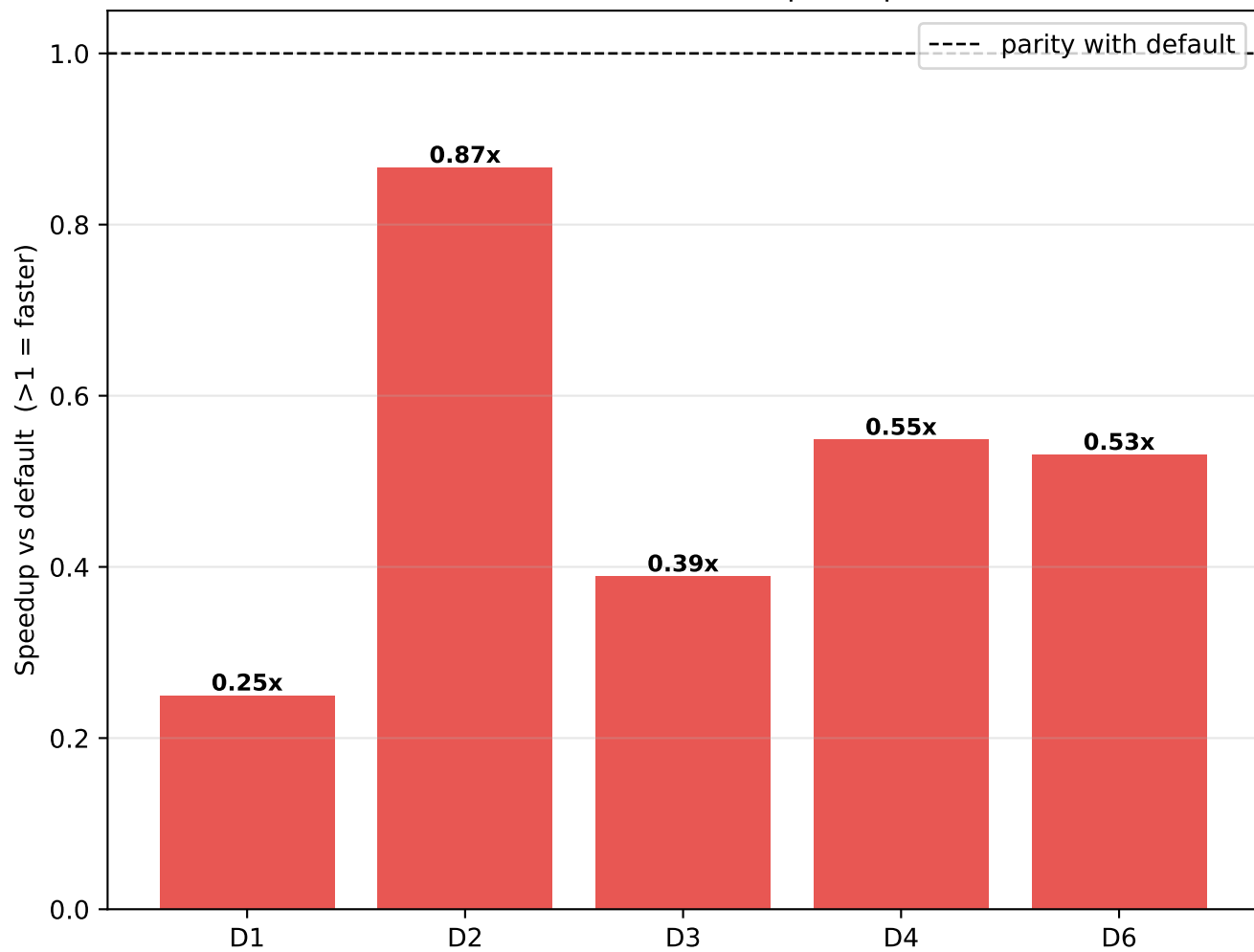
Wilcoxon on D6: no ground truth available (reads.fasta missing). Speedup vs default = 0.53x

Wilcoxon performance across datasets

F1: Wilcoxon vs default

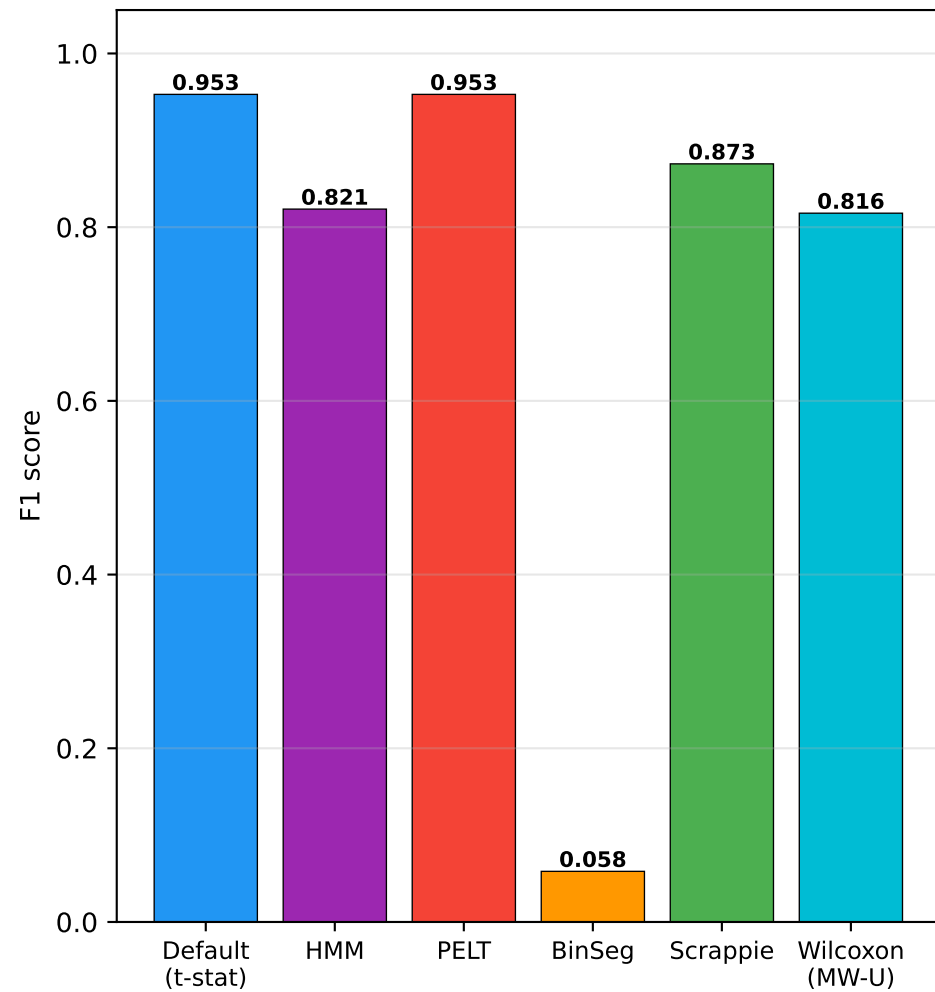


Wilcoxon wall-clock speedup

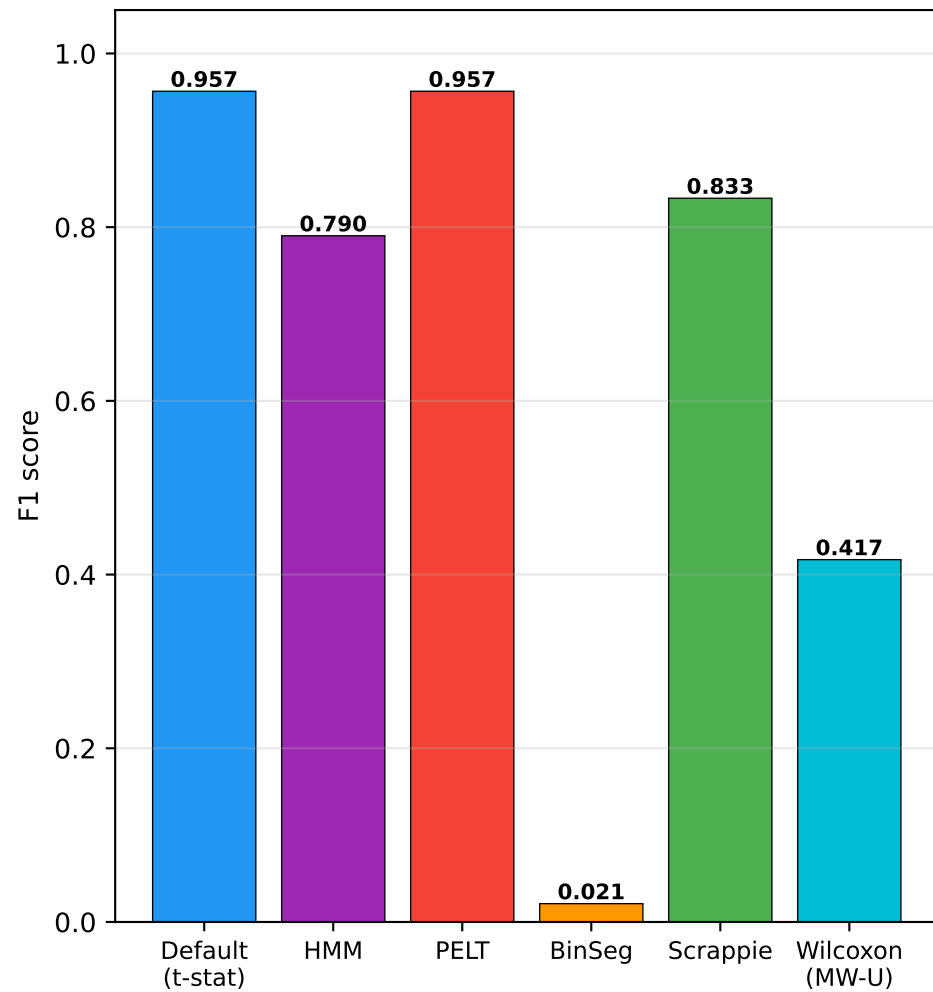


F1 across all six segmenters (datasets with ground truth)

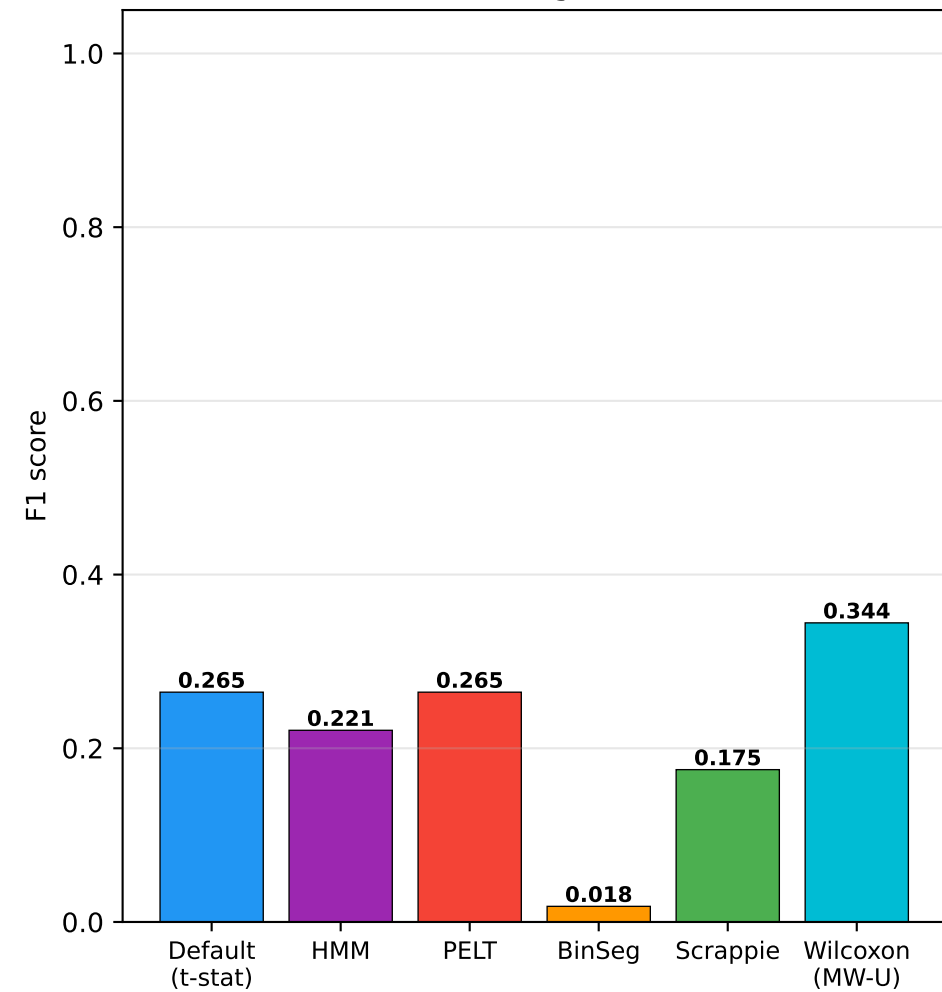
D1: SARS-CoV-2 (R9.4)



D3: Yeast (R9.4)

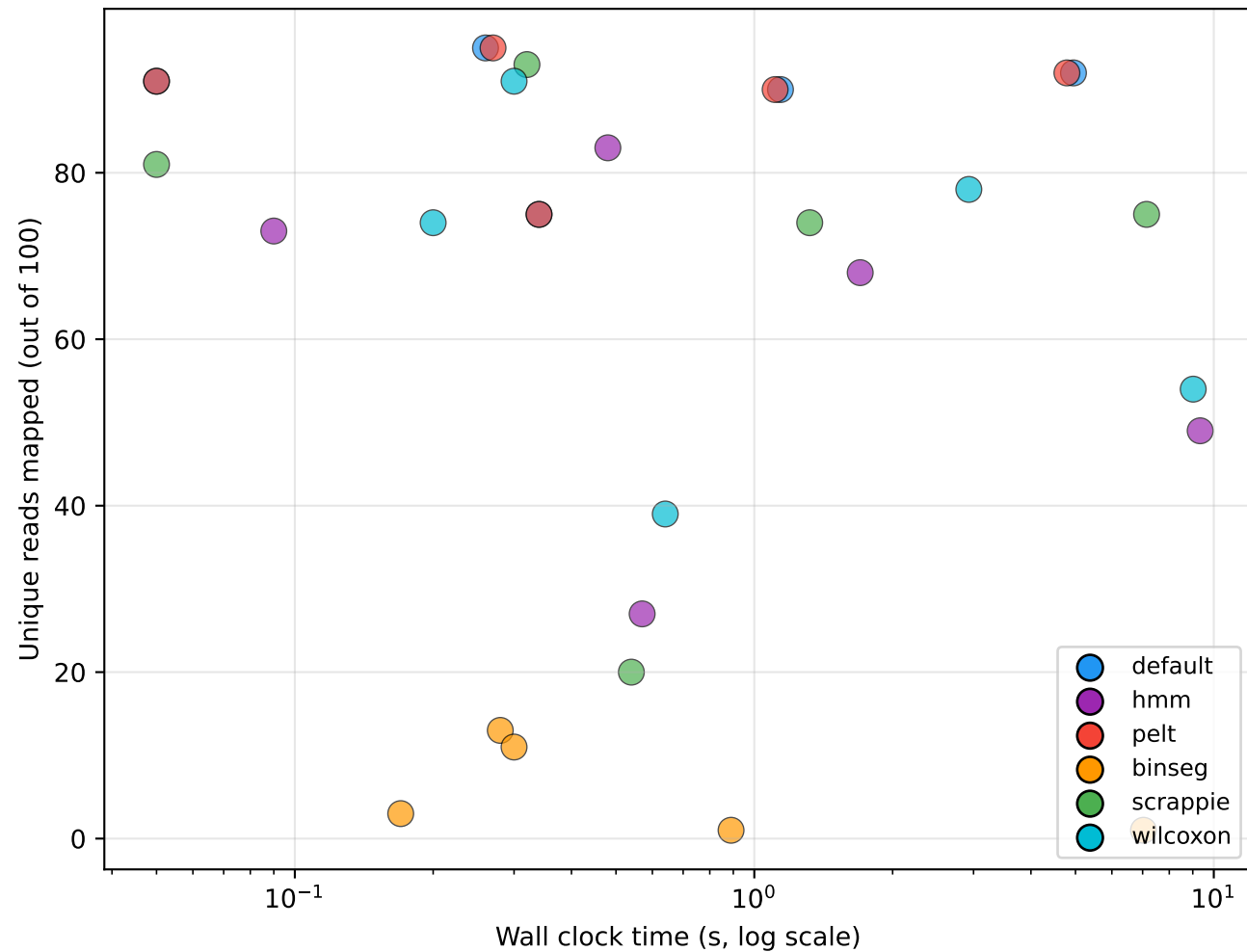


D4: Green Algae (R9.4)

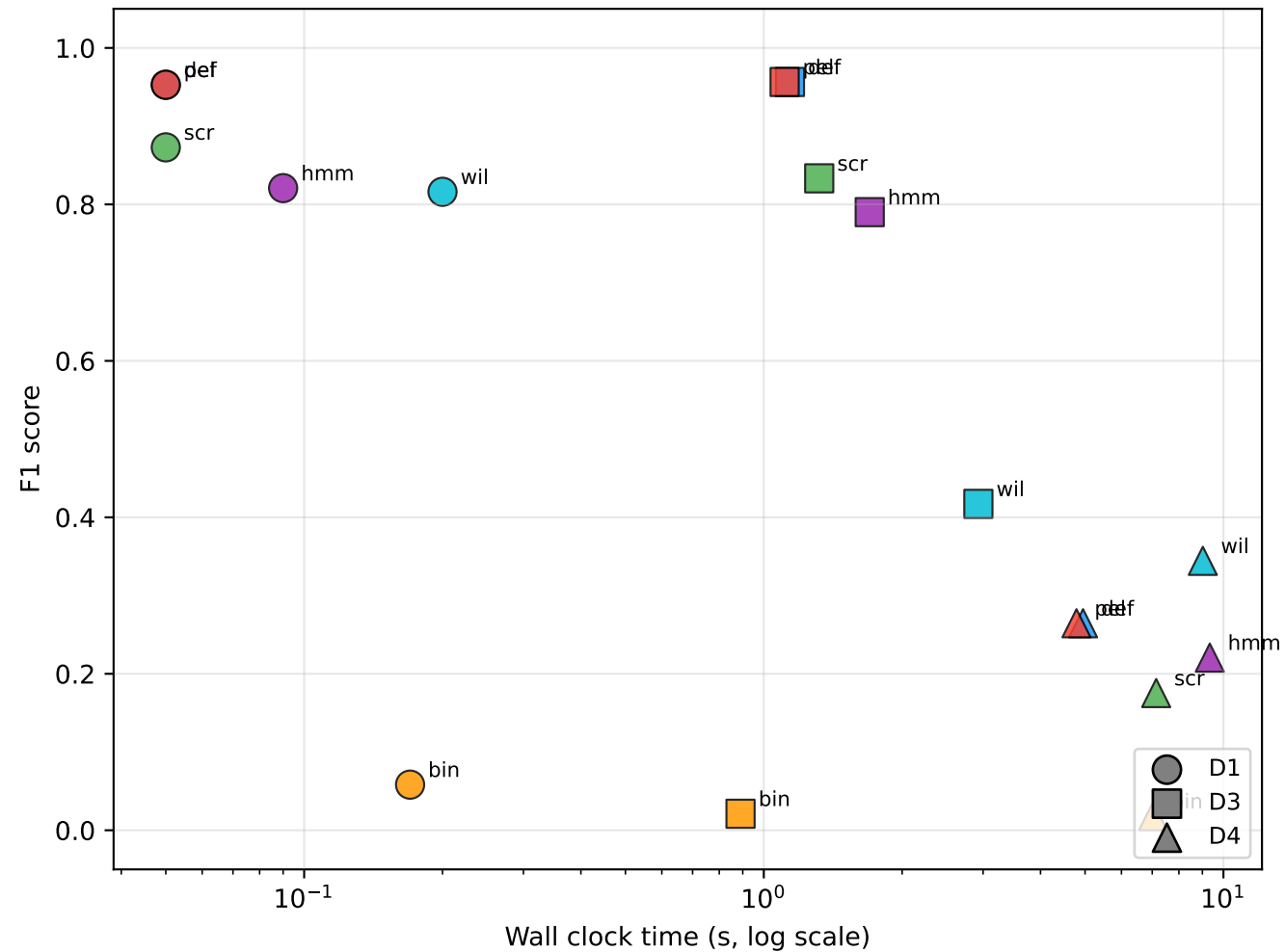


Speed vs Accuracy

All datasets: reads mapped vs time



F1 vs speed (datasets with ground truth)



Conclusions & V5 Findings

WILCOXON SEGMENTER -- DATASET-LEVEL RESULTS

Per-dataset F1 (where ground truth available):

D1	F1 = 0.816	(default = 0.953, delta = -0.137)
D3	F1 = 0.417	(default = 0.957, delta = -0.539)
D4	F1 = 0.344	(default = 0.265, delta = +0.080)

Per-dataset speedup vs default (>1 = wilcoxon faster):

D1	speedup = 0.25x
D2	speedup = 0.87x
D3	speedup = 0.39x
D4	speedup = 0.55x
D6	speedup = 0.53x

KEY FINDINGS

1. Wilcoxon is competitive on hard genomes, weaker on easy ones.
On D1 (SARS-CoV-2, 30 kb): F1 = 0.816 vs default 0.953 -- worse.
On D3 (yeast, 12 Mb): F1 = 0.417 vs default 0.957 -- much worse.
On D4 (algae, 111 Mb): F1 = 0.344 vs default 0.265 -- WINS by +0.08
D4 is the hardest dataset (most chromosomes, lowest absolute F1 across all segmenters); the rank-based statistic seems to handle the noise floor / outlier regime there better than the t-statistic.
2. Wilcoxon is slower than default on every dataset (speedup < 1x).
Range: 0.25x (D1) to 0.87x (D2). Pairwise $O(n \cdot w^2)$ inner loop costs ~2-4x of the t-stat on the segmentation step, but mapping wall-time is dominated by indexing + chaining + I/O so the end-to-end gap stays in the 1.1-4x range, never above parity.
3. Default/PELT win on D1, D2, D3; Wilcoxon wins on D4.
No segmenter dominates everywhere. Choice depends on signal regime: stationary Gaussian -> t-stat; heavy-tailed / drift -> Wilcoxon.
4. BinSeg still broken (F1 ~ 0.02-0.06) -- unchanged from V4.

FUTURE WORK

- * Per-dataset Wilcoxon hyperparameter sweep on real signals (the synth-derived defaults are not optimal for real nanopore data).
- * Try a robust-scaler-normalized Wilcoxon (median+MAD per window) to eliminate slow drift before ranking.
- * Complete D2/D6 ground truth once SRA reads.fasta retrievals finish.
- * Compare event count distributions: does Wilcoxon under-segment in no-event regions and over-segment in transitions?